

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY, BHOPAL

ELECTRONICS LAB

EXPERIMENT NO.1

CHARACTERISTICS OF OPERATIONAL AMPLIFIER

Aim: To study the Input Offset Current, Input & Output Offset Voltage of an operational Amplifier IC 741. To study the Gain Error, Band Width, Slew Rate & Output Impedance.

Instruments required:

The instrument comprises of the following built in parts:

1. Fixed Output DC Regulated Power Supply of $\pm 15V$
2. DC Regulated, Continuously Variable, Short Circuit & Overload protected Power Supply of 0-2V.
3. Operational Amplifier IC741 is placed inside the cabinet and connections brought out on sockets.
4. Combination of resistances are soldered behind the front panel.
5. Potentiometers are also provided on the front panel

Theory:

INPUT OFFSET CURRENT CHARACTERISTIC:

Input offset current is the difference of two input (inverting and noninverting) Currents. It tells us approximately what each input current is. Smaller the input offset current, smaller the possible unbalance.

Procedure

1. Connect the circuit as shown in Fig. No. (1a).
2. The DC voltages we are about to measure are in the millivolt range. One convenient way to measure these voltages is with an multimeter (millivoltmeter).
3. Connect multimeter between inverting input of operational amplifier and ground point. Record the DC voltage in Table No. (1b).

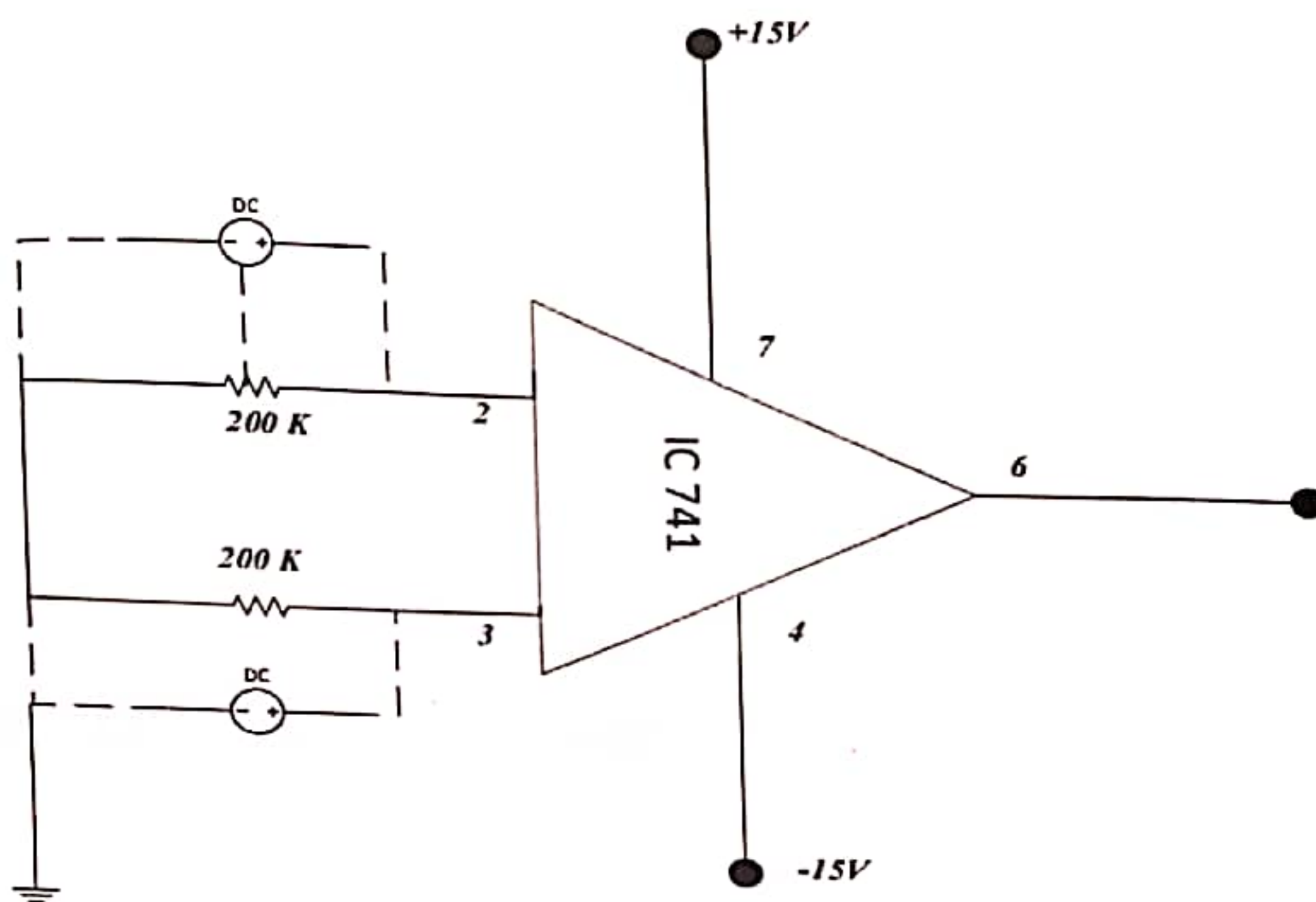


Fig. 1

INPUT MODE	DC VOLTAGE (mV)	INPUT OFFSET CURRENT CURRENT= DC VOLTAGE/200Kohm
INVERTING		
NON-INVERTING		

Table no. 1

4. Connect the multimeter between the noninverting input and the ground point. Also, record the DC voltage in Table No. (1).
5. Using Ohm's law calculate the input currents for inverting & non inverting inputs. The difference of these two currents will show the input offset current of operational amplifier.

$$\text{INPUT OFFSET CURRENT} = \text{NON-INV. CURRENT} - \text{INV. CURRENT}$$

PROCEDURE FOR CALCULATIONS OF INPUT & OUTPUT OFFSET VOLTAGES:

1. Connect the circuit as shown in Fig. No. (2a). Select the range of voltmeter to 2Volts.
2. Switch ON the instrument using ON/OFF toggle switch provided on front panel.
3. Measure the DC output offset voltage at pin 6 and record the result in Table 2(c).

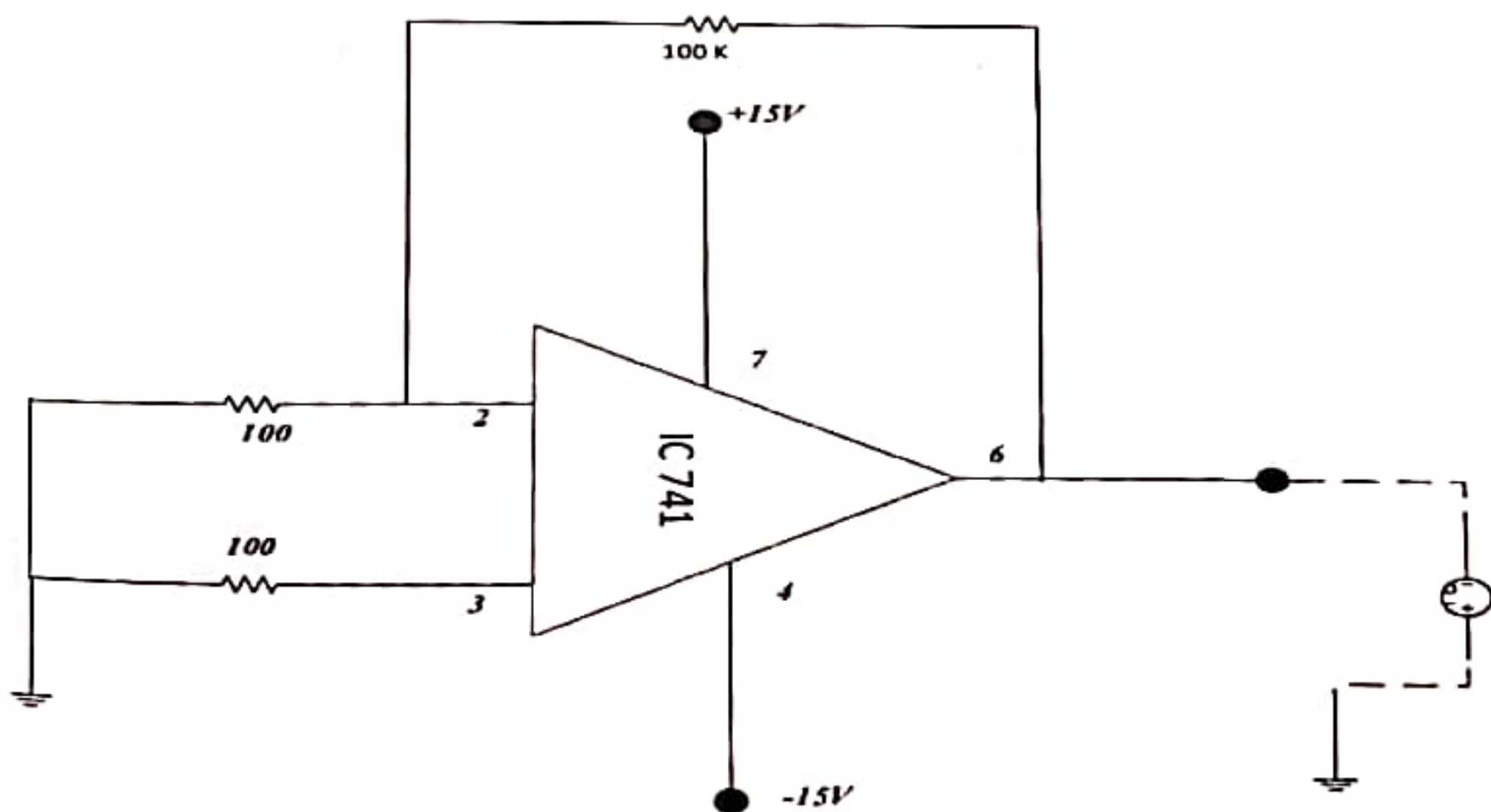


Fig. 2a

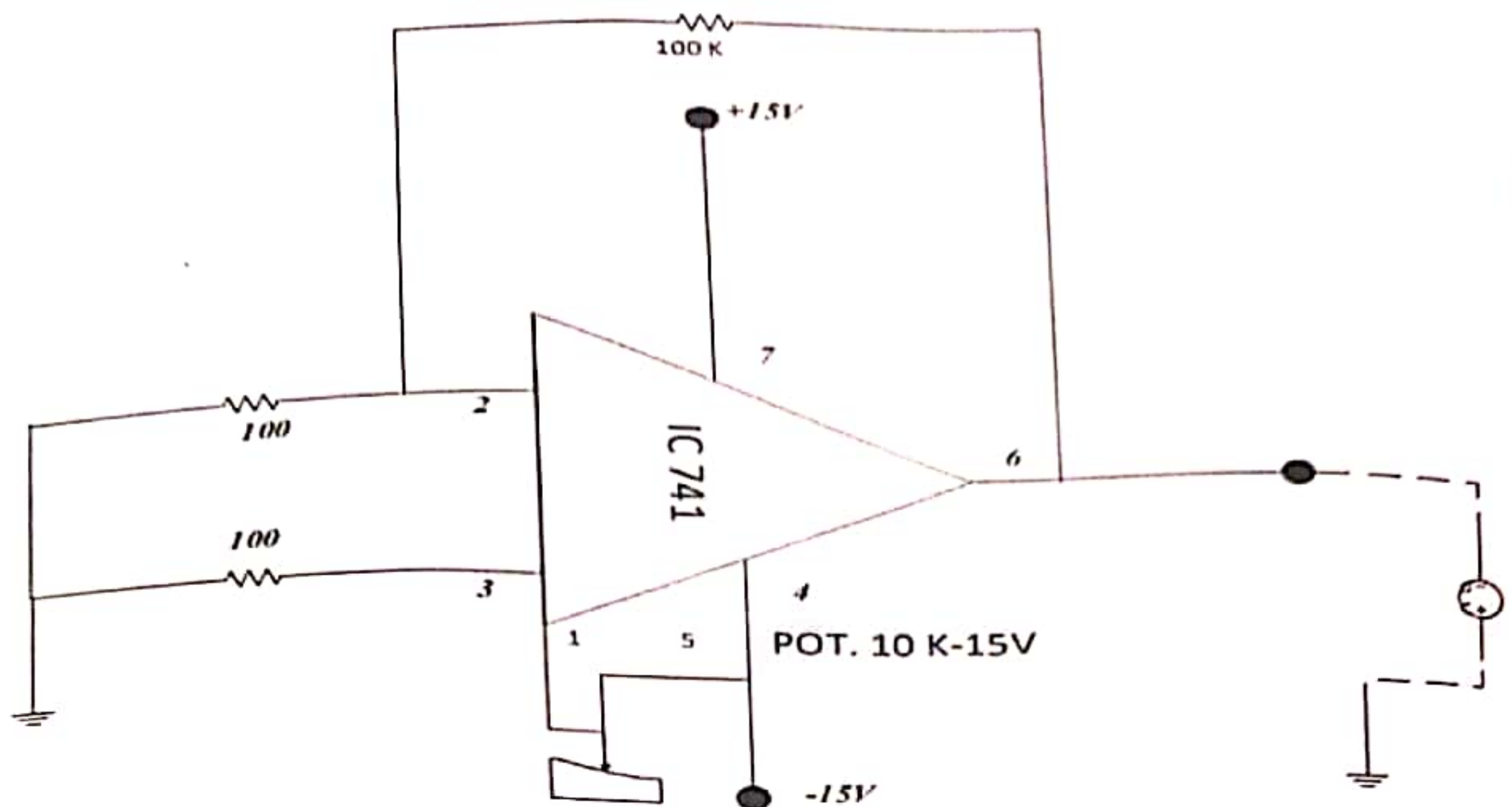


Fig. 2b

4. The voltage gain is approximately equal to the ratio of the feedback resistor to the input resistor. According to Fig. No. (2a), this means the voltage gain is approximately 1000, with the output voltage of Table (2c) calculate the input offset voltage using formula.

$$V_{in} = \frac{V_{out}}{1000}$$

Record this input offset voltage in Table (2c).

Vout	Vin

Table 2c

NULL BALACING:-

5. Switch OFF the instrument and add a 10k potentiometer in the circuit as shown in Fig. No. (2b). Change the range of voltmeter to 20 Volts and switch ON the instrument again. Note down the output voltage at pin 6. Adjust the potentiometer until the output offset voltage is zero.

GAIN ERROR:-

An operational amplifier is a high gain, direct coupled differential amplifier whose gain is controlled by external negative feedback circuitry. When we apply a large amplitude signal at the input of the op-amp the gain of op-amp is distorted (output wave shape distorted). this is known as the gain error of the op-amp. Note that particular input signal and the frequency.

Procedure

1. Connect the circuit as shown in Fig. No. (3).

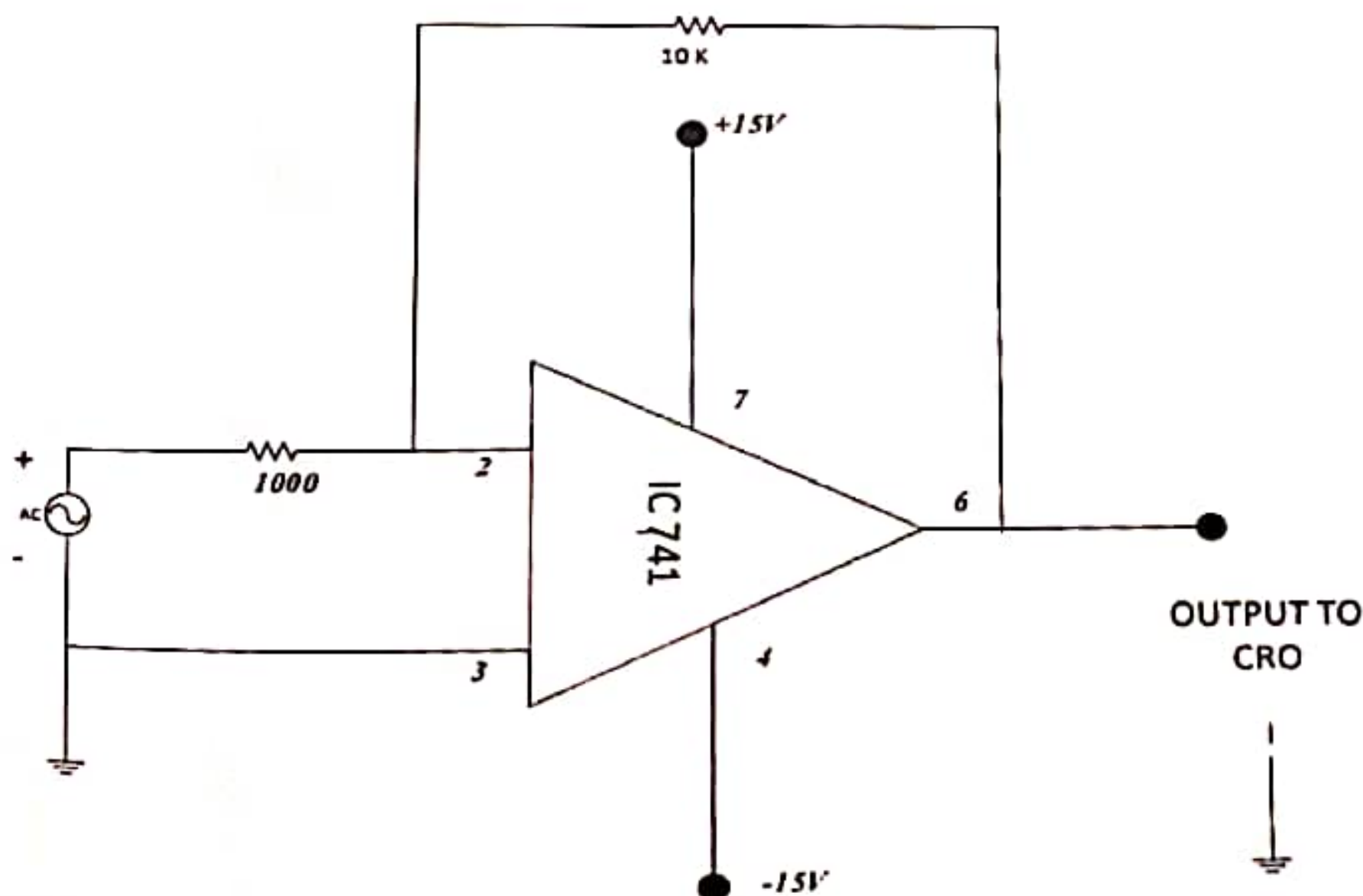


Fig. 3

2. Set the function generator at 1KHz frequency, 0.2V amplitude (P-P). Also connect CRO at output terminals.
3. Switch ON the instrument using ON/OFF toggle switch provided in the front panel.
4. Gradually increase the output amplitude from the function generator just below the point where the waveform distorts. Measure and record the peak to peak output signal voltage and also input signal voltage. This is the maximum undistorted output signal for the feedback resistors in the circuits.

Gain of op-amp = V_{out}/V_{in}

5. Now increase the output from function generator just above the point where the wave form distorts. Measure and record the peak to peak input signal and output distorted signal.

POWER BANDWIDTH:-

Slew rate distortion of a sine wave starts at a point where the slope of the sine wave equals the slew rate of the Operational Amplifier with advanced mathematics, it is possible to derive this useful formula.

$$F_{max} = S_n / 2\pi \cdot V_p$$

Where F_{max} = highest undistorted frequency.

S_n is slew rate of Op-Amp.

V_p is peak voltage of sine wave.

e.g. if the output sine wave has a peak voltage of 10 Volt & the slew rate of 0.5 Volt per micro seconds, the maximum frequency for large signal operation is $f_{max} = 0.5 \text{ V/microsecond} / 2\pi \cdot 10 \text{ V} = 7.9 \text{ KHz}$

Frequency f_{max} is called the Power Bandwidth of an Op-Amp. We have just found the 8KHz frequency for a signal of 10 Volt Peak. This means the undistorted Band Width for large signal operation is 8 KHz. Try to amplify higher frequency of the same peak value you will get Slew rate distortion.

1. Connect the circuit as shown in Fig. No. (4a).

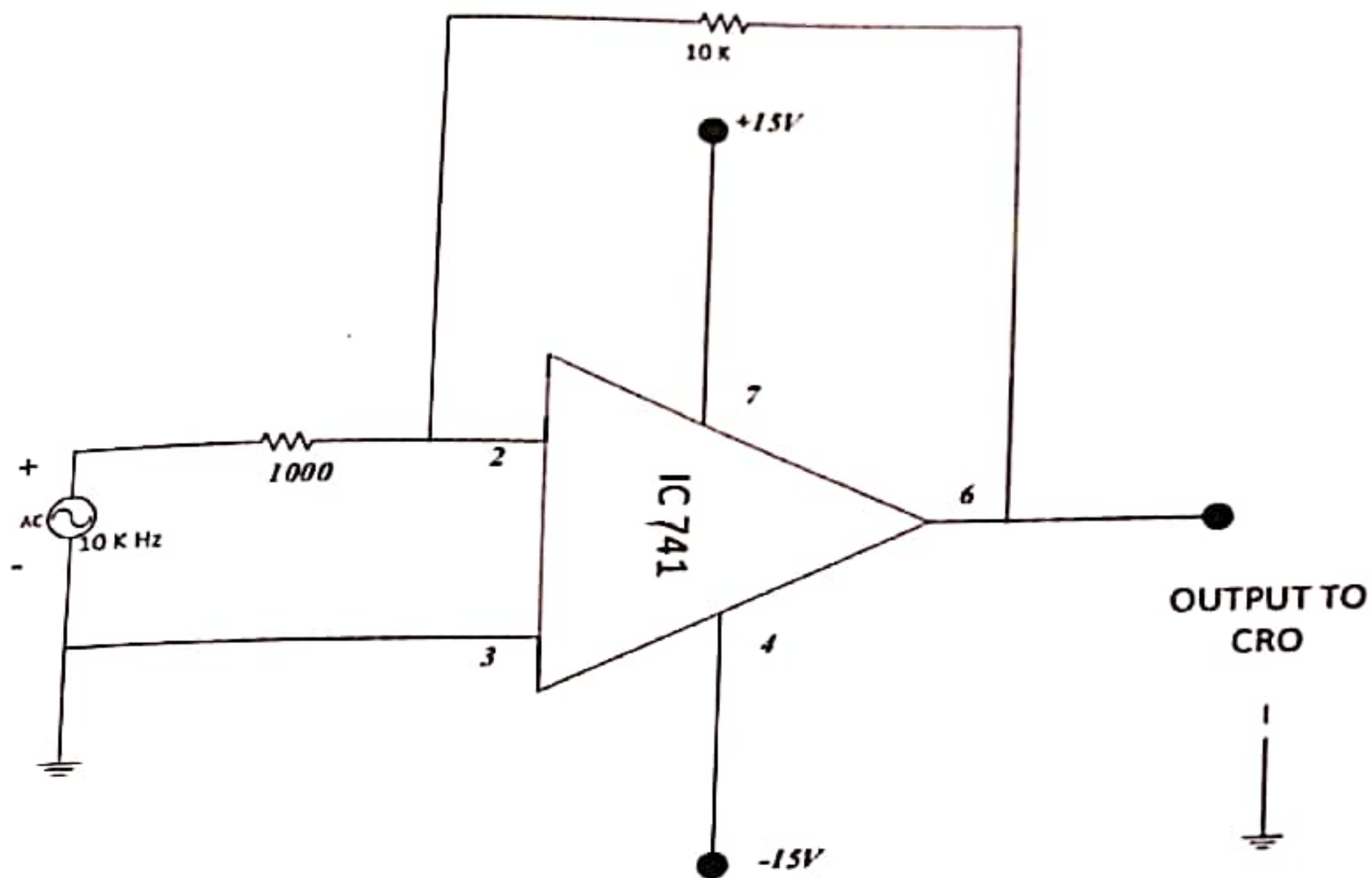


Fig 4a

2. Set the Function Generator output at 1 KHz & adjust the signal level to get 20V peak to peak at the output of Operational Amplifier.
3. Increase the frequency & observe the wave form at the CRO.
4. Somewhere nearer to 10 KHz slew rate distortion will become evident because the wave form will appear Triangular & the amplitude will decrease.

TO MEASURE OUTPUT IMPEDANCE:

1. Connect the circuit as shown in Fig. No. (5). Select the range of voltmeter to 20 Volts.

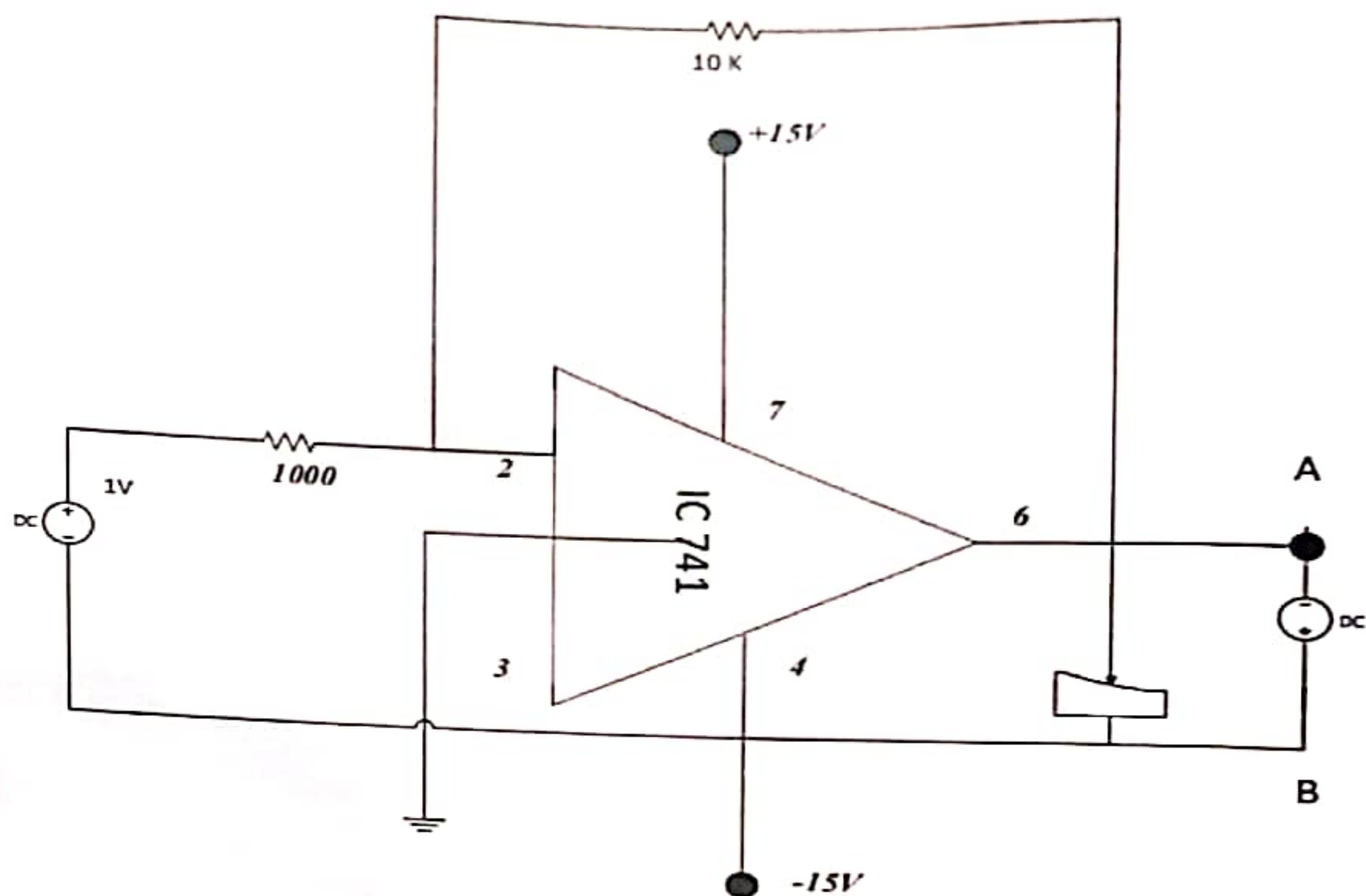


Fig 5

2. Switch ON the instrument using ON/OFF toggle switch provided on the front panel
3. Apply IVDC at the input of circuit & note down the output voltage from voltmeter.
4. To measure output impedance, connect potentiometer (1k) provided on the front panel across point 'A' & 'B' ie, across output. Vary the value of potentiometer resistance, so that output voltage will remains half of the voltage noted in step (3).
5. Disconnect the potentiometer from output and note down the value of resistance using a multimeter. This resistance is the output impedance of the circuit.

Result & Conclusion:

Precautions:

1. Before assembling or testing a circuit, thoroughly understand the circuit diagram and the components involved. This ensures correct wiring and component placement.
2. Ensure that the components you use (resistors, capacitors, ICs, etc.) match the values specified in the circuit design to avoid malfunction or damage.
3. Always turn off or disconnect the power supply when making changes to the circuit. This reduces the risk of electric shock, short circuits, or component damage.
4. Some components (like resistors and transistors) can overheat if the circuit is not designed properly or if there is a fault. Always check for excessive heat and turn off the power if needed.
5. When using ICs, check the pin 1 marking and ensure the IC is inserted with correct orientation to avoid shorting pins or damaging the IC.
6. Be gentle when inserting or removing components from the breadboard or PCB. Use appropriate tools like tweezers to handle small or delicate parts.

Questions:

1. How can you minimize the effect of the input offset current in a practical circuit?
2. How does the input offset voltage affect the precision of the output signal?
3. How does the output offset voltage vary with temperature changes in the IC?
4. How can you minimize gain error in high-precision circuits?
5. How does temperature affect the gain error in IC 741?
6. How does the bandwidth change as the gain increases in IC 741?
7. How can you compensate for a low slew rate in a system?
8. How does the output impedance influence the ability of IC 741 to drive loads with varying impedance?